

Singular Perturbations, Moments and Descriptor Systems: Linear

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1 Linear Systems - Thinking about explicit solutions and moments

Consider the following linear time-invariant, SISO system:

$$\begin{aligned}\dot{x} &= Ax + Bu, \\ y &= Cx,\end{aligned}\tag{1}$$

$$\begin{aligned}\dot{w} &= Sw, \\ u &= Lw,\end{aligned}\tag{2}$$

Given initial conditions $x(0) = x_0$ and $w(0) = w_0$, the solution to the system is given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}BL e^{S\tau}w_0 d\tau.\tag{3}$$

The output of the system is given by

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}BL e^{S\tau}w_0 d\tau.\tag{4}$$

If $\sigma(A) \cup \sigma(S) = \emptyset$, then there exists a unique solution Π to the Sylvester equation

$$A\Pi + BL = \Pi S.\tag{5}$$

Using this, we can substitute an expression for BL into the integral term for $x(t)$:

$$\begin{aligned}x(t) &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}(\Pi S - A\Pi)e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}\Pi S e^{S\tau}w_0 d\tau - \int_0^t e^{A(t-\tau)}A\Pi e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + (\Pi e^{St} - e^{At}\Pi)w_0.\end{aligned}\tag{6}$$

Forward Moments are defined to be

$$M_g = C\Pi\tag{7}$$

2 Problem Setup

Consider the following linear singularly perturbed system:

$$\begin{aligned}\dot{x}_1(t) &= A_{11}x_1(t) + A_{12}x_2(t) + B_1u(t), \\ \varepsilon\dot{x}_2(t) &= A_{21}x_1(t) + A_{22}x_2(t) + B_2u(t),\end{aligned}\tag{8}$$

where $x_1(t) \in \mathbb{R}^{n_1}$, $x_2(t) \in \mathbb{R}^{n_2}$, $u(t) \in \mathbb{R}^m$, and $0 < \varepsilon \ll 1$.

You can write this as a descriptor system:

$$E_1\dot{x} = Ax + Bu\tag{9}$$

where

$$E_1 = \begin{bmatrix} I_{n_1} & 0 \\ 0 & \varepsilon I_{n_2} \end{bmatrix}, \quad A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \quad (10)$$

The input signal is also a linear singularly perturbed system:

$$\begin{aligned} \dot{w}_1 &= S_{11}w_1 + S_{12}w_2 \\ \varepsilon \dot{w}_2 &= S_{21}w_1 + S_{22}w_2 \end{aligned} \quad (11)$$

where $w_1(t) \in \mathbb{R}^{r_1}$, $w_2(t) \in \mathbb{R}^{r_2}$, and $0 < \varepsilon \ll 1$. The input to the system is given by $u(t) = L_1w_1(t) + L_2w_2(t)$.

The input signal can also be written as a descriptor system like so:

$$E_2\dot{w} = Sw \quad (12)$$

where

$$E_2 = \begin{bmatrix} I_{r_1} & 0 \\ 0 & \varepsilon I_{r_2} \end{bmatrix}, \quad S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}. \quad (13)$$

The output of the system is given by

$$y(t) = C_1x_1(t) + C_2x_2(t). \quad (14)$$

3 Wong Sequences

Stuff on Wong Sequences

4 Wong Sequences: Regular

We now assume that the pencils (E_1, A) and (E_2, S) are regular. Then as shown in the summary by Stephan Trenn, Theorem 2.12:

$$P(sE_1 - A)Q = s \begin{bmatrix} I & 0 \\ 0 & N_1 \end{bmatrix} - \begin{bmatrix} J_1 & 0 \\ 0 & I \end{bmatrix}. \quad (15)$$

$$W(sE_2 - S)T = s \begin{bmatrix} I & 0 \\ 0 & N_2 \end{bmatrix} - \begin{bmatrix} J_2 & 0 \\ 0 & I \end{bmatrix}. \quad (16)$$

The following quantities can then be defined for both the system and the signal. The index for system is 1, and signal quantities have index 2. The consistency projector is:

$$\Xi_1 = Q \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \quad (17)$$

The differential projector:

$$\Theta_1 = Q \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} P \quad (18)$$

The consistency projector:

$$\Phi_1 = Q \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} P \quad (19)$$

Furthermore:

$$\tilde{A} := \Theta_1 A \quad (20)$$

$$\hat{E} := \Phi_1 E_1 \quad (21)$$

There are some conditions on (P, Q) related to the Wong sequences. The explicit solution can then be found as the following: Solution for the signal:

$$w(t) = e^{\tilde{S}t} \Xi_2 c_2 \quad (22)$$

The input is then given by:

$$f(t) = BLw(t) = BLE^{\tilde{S}t}\Xi_2c_2 \quad (23)$$

$$x(t) = e^{\tilde{A}t}\Xi_1c + \int_0^t e^{\tilde{A}(t-\tau)}\Xi_1BLE^{\tilde{S}\tau}\Xi_2c_2d\tau - \sum_{i=0}^{n-1} \hat{E}^i\Phi_1\left(BLe^{\tilde{S}t}\Xi_2c_2\right)^{(i)} \quad (24)$$

There are some conditions on c_1, c_2 relating the initial conditions.

We now investigate the integral:

$$\int_0^t e^{\tilde{A}(t-\tau)}\Xi_1BLE^{\tilde{S}\tau}\Xi_2c_2d\tau = e^{\tilde{A}t} \left(\int_0^t e^{-\tilde{A}\tau}\Xi_1BLE^{\tilde{S}\tau}d\tau \right) \Xi_2c_2 \quad (25)$$

If $\sigma(\tilde{A}) \cup \sigma\tilde{S} = \emptyset$:

$$\tilde{A}\Pi + \Xi_1BL = \Pi\tilde{S} \quad (26)$$

Then the integral evaluates to:

$$\int_0^t e^{\tilde{A}(t-\tau)}\Xi_1BLE^{\tilde{S}\tau}\Xi_2c_2d\tau = \left(\Pi e^{\tilde{S}t} - e^{\tilde{A}t}\Pi \right) \Xi_2c_2 \quad (27)$$

This is analagous to the result shown in section 1.